Section 4: Ecology and the environment

a) The organism in the environment

b) Feeding relationships

c) Cycles within ecosystems

d) Human influences on the environment

a) The organism in the environment

Students will be assessed on their ability to:

4.1 understand the terms population, community, habitat and ecosystem

4.2 explain how quadrats can be used to estimate the population size of an organism in two different areas

4.3 explain how quadrats can be used to sample the distribution of organisms in their habitats.

b) Feeding relationships

Students will be assessed on their ability to:

4.4 explain the names given to different trophic levels to include producers, primary, secondary and tertiary consumers and decomposers

4.5 understand the concepts of food chains, food webs, pyramids of number, pyramids of biomass and pyramids of energy transfer

4.6 understand the transfer of substances and of energy along a food chain

4.7 explain why only about 10% of energy is transferred from one trophic level to the next.

c) Cycles within ecosystems

Students will be assessed on their ability to:

4.8 describe the stages in the water cycle, including evaporation, transpiration, condensation and precipitation

4.9 describe the stages in the carbon cycle, including respiration, photosynthesis, decomposition and combustion

4.10 describe the stages in the nitrogen cycle, including the roles of nitrogen fixing bacteria, decomposers, nitrifying bacteria and denitrifying bacteria (specific names of bacteria are not required).

d) Human influences on the environment

Students will be assessed on their ability to:

4.11 understand the biological consequences of pollution of air by sulfur dioxide and by carbon monoxide

4.12 understand that water vapour, carbon dioxide, nitrous oxide, methane and CFCs are greenhouse gases

4.13 understand how human activities contribute to greenhouse gases

4.14 understand how an increase in greenhouse gases results in an enhanced greenhouse effect and that this may lead to global warming and its consequences

4.15 understand the biological consequences of pollution of water by sewage, including increases in the number of micro-organisms causing depletion of oxygen

4.16 understand that eutrophication can result from leached minerals from fertiliser

4.17 understand the effects of deforestation, including leaching, soil erosion, disturbance of the water cycle and of the balance in atmospheric oxygen and carbon dioxide.